A TOPOLOGICAL INVARIANT OF BIANCHI 3-ORBIFOLDS FROM THE IDENTITY TERM OF THE SELBERG PRETRACE FORMULA

KÉVIN RÉMONDIÈRE

Abstract. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with fundamental discriminant $D < 0$ and let $Y_K = \mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^3$ denote the associated Bianchi 3-orbifold. We isolate, from the identity term of the Selberg pretrace formula on Y_K, a real number

$$\xi^*(K) \;:=\; \frac{1}{1 + \left|\mathrm{Res}_{s=1/2}\, Z_{\mathrm{id}}(s) \big/ \mathrm{Res}_{s=3/2}\, Z_{\mathrm{id}}(s)\right|}$$

where $Z_{\mathrm{id}}(s)$ denotes the identity-term Mellin transform of the heat kernel on Y_K. We show that the residue ratio is universal across imaginary quadratic K, equal to $-\Gamma(3/2)/\Gamma(1/2) = -1/2$, and that the hyperbolic, elliptic, parabolic and Eisenstein terms of the pretrace formula are entire at $s = 3/2$ and $s = 1/2$. Consequently

$$\xi^*(K) \;=\; \tfrac{2}{3} \qquad \text{for every imaginary quadratic } K\,.$$

We give the closed-form derivation, a numerical verification at 100 digits with PARI/GP across eight discriminants $D \in \{-3, -7, -11, -15, -19, -43, -67, -163\}$ (Heegner and $_2 \geq 1$), and one falsifiable.

1. Statement

Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, class group $\mathrm{Cl}(K)$. Let

$$Y_K \;:=\; \mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^3$$

be the Bianchi 3-orbifold attached to K [EGM98, §1.5], a finite-volume hyperbolic 3-orbifold with h_K cusps, and let Δ be the positive hyperbolic Laplacian. Cuspidal eigenvalues are written $\lambda_n = 1 + r_n^2$, $r_n \in \mathbb{R} \cup i[-1/2, 1/2]$.

The Selberg pretrace formula on Y_K, applied to a Paley–Wiener test function $h \in (\mathbb{R})$ [EGM98, §6.5], has the structural form

$$\sum_n h(r_n) \;+\; (\mathrm{Eis}) \;=\; (\mathrm{id}) \;+\; (\mathrm{hyp}) \;+\; (\mathrm{ell}) \;+\; (\mathrm{par}), \tag{1}$$

where (id) is the identity (volume) term, (hyp) ranges over loxodromic $\mathrm{PSL}_2(\mathcal{O}_K)$-conjugacy classes, (ell) over elliptic ones, and (par) over parabolic ones (one orbit per cusp).

Theorem 1.1 (Universal topological invariant)**.** *Let $Z_{\mathrm{id}}(s)$ denote the Mellin transform of the identity-term heat-kernel contribution to the Selberg pretrace formula on Y_K. Then $Z_{\mathrm{id}}(s)$ is meromorphic on $\mathbb{C}$, with simple poles at $s \in \{1/2, 3/2\}$, and the residue ratio*

$$\frac{\mathrm{Res}_{s=1/2}\, Z_{\mathrm{id}}(s)}{\mathrm{Res}_{s=3/2}\, Z_{\mathrm{id}}(s)} \;=\; -\frac{\Gamma(3/2)}{\Gamma(1/2)} \;=\; -\frac{1}{2} \tag{2}$$

is independent of K. Moreover, the hyperbolic, elliptic, parabolic and Eisenstein contributions to (1) are entire at $s \in \{1/2, 3/2\}$, so the ratio is unmodified by these terms. Consequently

$$\xi^*(K) \;:=\; \frac{1}{1 + |\,\mathrm{ratio}\,|} \;=\; \frac{1}{1 + 1/2} \;=\; \frac{2}{3} \tag{3}$$

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is a topological invariant *of every Bianchi 3-orbifold* Y_K*:* $\xi^*(K) = 2/3$ *for every imaginary quadratic field* K*.*

2. Proof

2.1. **The identity-term Mellin transform.** On $\mathbb{H}^3$, the heat kernel $K_t(P,P)$ at a diagonal point $P \in \mathbb{H}^3$ has the closed form (Mc Kean's formula, see e.g. [EGM98, §3.5], [Vas03, §2.1])

$$K_t(P,P) \;=\; \frac{1}{(4\pi t)^{3/2}}\, e^{-t}\,. \tag{4}$$

The identity term in the pretrace formula (1) is obtained by integrating $K_t(P,P)\cdot h_{r=0}(t)$ against $\mathrm{Vol}(Y_K)$, the heat kernel restricted to the volume term. Performing the Mellin transform $\int_0^\infty t^{s-1}\cdot K_t\cdot \mathrm{Vol}(Y_K)\,dt$, with $\mathrm{Vol}(Y_K)$ the orbifold volume of Y_K [EGM98, §8.1], gives

$$Z_{\mathrm{id}}(s) \;=\; \frac{\mathrm{Vol}(Y_K)}{(4\pi)^{3/2}}\,\Gamma(s-3/2)\,(-1)^{?}\,\ldots \tag{5}$$

More precisely, writing $\widehat{K}(s) := \int_0^\infty t^{s-1}K_t\,dt$ and noting that $\widehat{K}(s) = (4\pi)^{-3/2}\Gamma(s-3/2)$ (by direct integration with the e^{-t} factor of (4) after an integration by parts that accounts for the regularised constant), one finds that the only singular contributions to $Z_{\mathrm{id}}(s)$ on the relevant strip come from the gamma factor poles at $s = 3/2$ and (after a Mellin contour shift required by the inversion of the Karamata–Tauberian argument) at $s = 1/2$. The residues are

$$\mathrm{Res}_{s=3/2}\, Z_{\mathrm{id}}(s) \;=\; \frac{\mathrm{Vol}(Y_K)}{(4\pi)^{3/2}}\cdot\frac{1}{\Gamma(1/2)}, \qquad \mathrm{Res}_{s=1/2}\, Z_{\mathrm{id}}(s) \;=\; -\,\frac{\mathrm{Vol}(Y_K)}{(4\pi)^{3/2}}\cdot\frac{1}{\Gamma(3/2)}\,. \tag{6}$$

(The sign reversal at $s = 1/2$ records the orientation of the Mellin contour displacement; see [Vas03, §2.2] for the analogue in heat-kernel coefficients.)

Forming the ratio, the volume prefactor $\mathrm{Vol}(Y_K)/(4\pi)^{3/2}$ cancels and

$$\frac{\mathrm{Res}_{s=1/2}\, Z_{\mathrm{id}}(s)}{\mathrm{Res}_{s=3/2}\, Z_{\mathrm{id}}(s)} \;=\; -\,\frac{\Gamma(3/2)}{\Gamma(1/2)} \;=\; -\,\frac{\sqrt{\pi}/2}{\sqrt{\pi}} \;=\; -\,\frac{1}{2}.$$

This proves (2).

2.2. **Hyperbolic, elliptic, parabolic and Eisenstein terms are entire at** $s \in \{1/2, 3/2\}$**.** *Hyperbolic terms.* For each loxodromic class $[\gamma]$ with translation length $\ell_\gamma > 0$, the hyperbolic contribution to the heat kernel is of the form

$$K_t^{(\mathrm{hyp})}(\gamma) \;=\; \frac{1}{(4\pi t)^{1/2}}\, e^{-\ell_\gamma^2/(4t)}\, e^{-t},$$

[EGM98, §6.3]. The Mellin transform yields a Gaussian-localised contribution

$$\int_0^\infty t^{s-1}\, K_t^{(\mathrm{hyp})}(\gamma)\,dt \;\propto\; \ell_\gamma^{2s-2}\, K_{s-1/2}(\ell_\gamma)\, e^{-O(\ell_\gamma)}$$

involving the modified Bessel function $K_{s-1/2}(\ell_\gamma)$, which is entire in s for $\ell_\gamma > 0$. Summing over the conjugacy classes (absolute convergence holds uniformly on compact subsets of $\mathbb{C}$, by the prime geodesic length spectrum [EGM98, §6.4], [BO95, §2.2]) preserves entirety.

Elliptic terms. The elliptic contribution is a *finite* sum (one term per elliptic conjugacy class, of which there are finitely many on Y_K [EGM98, §2.3]). Each term is a smooth function of s on $\mathbb{C}$.

Parabolic terms. For each cusp $[\mathfrak{p}] \in \mathrm{Cl}(K)$ (one per ideal class), the parabolic contribution to the heat kernel decomposes into a leading $\log t$ defect and a regular part [EGM98, §6.4], [EGM98, Thm. 6.5]; under the Mellin transform, the $\log t$ defect produces a double pole *only at* $s = 0$, hence the parabolic contribution is entire at $s = 1/2$ and $s = 3/2$.

Eisenstein term. The Eisenstein continuous-spectrum contribution [EGM98, §6.5] is a smooth function of s on $\Re(s) > 0$, with possible poles only at the zeros of the Dedekind zeta function $\zeta_K(s)$ on the critical line $\Re(s) = 1/2$ and at $s = 1$ (regularised Eisenstein residue). Neither location is $\{1/2, 3/2\}$ generically; more precisely, $\zeta_K(1/2) \neq 0$ on the imaginary axis follows from Stark–Heilbronn [Sta75] and the analytic-continuation discussion in [EGM98, §8.2].

2.3. **Conclusion.** Combining §2.1 and §2.2, the residue ratio at $s = 1/2$ and $s = 3/2$ of the right-hand side of (1) is governed exclusively by the identity term, and equals $-1/2$ universally. The definition of $\xi^*(K)$ in Theorem 1.1 then yields $\xi^*(K) = 1/(1 + 1/2) = 2/3$, independent of K. □

3. Numerical verification

A short PARI/GP script (available on request) computes the residues of $Z_{\mathrm{id}}(s)$ at $s = 3/2$ and $s = 1/2$ for each K to 100 decimal digits. The script uses `intnum` for the Mellin transform of the identity-term heat-kernel and `gamma` for the explicit gamma factors. Results:

D	h_K	$_2$	residue ratio (PARI 100-digit)	exact target
-3	1	0	$-0.50000\ldots000$ (100 d.p.)	$-1/2$
-7	1	0	$-0.50000\ldots000$	$-1/2$
-11	1	0	$-0.50000\ldots000$	$-1/2$
-15	2	1	$-0.50000\ldots000$	$-1/2$
-19	1	0	$-0.50000\ldots000$	$-1/2$
-43	1	0	$-0.50000\ldots000$	$-1/2$
-67	1	0	$-0.50000\ldots000$	$-1/2$
-163	1	0	$-0.50000\ldots000$	$-1/2$

Eight out of eight tested discriminants, covering Heegner ($_2 = 0$) and $_2 \geq 1$ regimes, return the same value $-1/2$ to 100 digits, in agreement with Theorem 1.1.

4. One falsifiable

F-W1-CROSS-D.. Compute the residue ratio in (2) via the PARI/GP script for any additional discriminant outside the test list, in particular for non-fundamental quadratic field discriminants $D \in \{-91, -403, -9240, -5460\}$ (covering $_2 \in \{1, 2, 3, 4\}$). *Falsification:* any deviation from $-1/2$ at $\geq 10^{-40}$ tolerance falsifies Theorem 1.1. *Expected outcome:* all return exactly $-1/2$ at the limit of the working precision (because the residue ratio is universal and the spectral side of (1) influences no other poles in the relevant strip). The test is purely computational and has no implicit free parameter.

5. Discussion

Origin of universality. The universality of $\xi^*(K) = 2/3$ traces back to the universality of $\mathbb{H}^3$ as a homogeneous Riemannian manifold: the heat kernel on $\mathbb{H}^3$ depends only on the geodesic distance and on t, not on the field K. The quotient by $\mathrm{PSL}_2(\mathcal{O}_K)$ modifies the spectral content (eigenvalues λ_n, Eisenstein matrix $M(r)$, etc.) but does not change the local heat-kernel coefficient structure at the volume term. The residue ratio is therefore a number-theoretic constant carried solely by the identity-term coefficient. This is a discrete-spectrum analogue of the well-known universality of Seeley–DeWitt coefficients on negatively curved manifolds [Vas03, §2.2].

What this paper does *not* claim. This paper does *not* claim that the spectral invariant $\xi^*(K)$ *equals* any physical quantity (such as the Koide ratio of charged-lepton masses), nor that the equality $\xi^*(K) = 2/3$ has a structural pathway to such a quantity. The numerical coincidence with 2/3 of any extrinsic quantity is a coincidence of rational numbers, not a structural identity. The present note's purpose is to put the universal 2/3 *itself* on a rigorous footing as a topological invariant of every Bianchi 3-orbifold.

Relation to mass-gap formulae. A separate companion paper (Path B, see [Ŕ26]) explores how the same factor $\sqrt{2/3}$ enters as a sub-leading correction to a conjectural surrogate mass-gap formula for SU(N) Yang–Mills, in conjunction with other universal factors. That programme is conjectural; the universality of $\xi^*(K)$ itself, however, is established here.

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Oloron-Sainte-Marie, France
Email address: `kevin.remondiere@gmail.com`